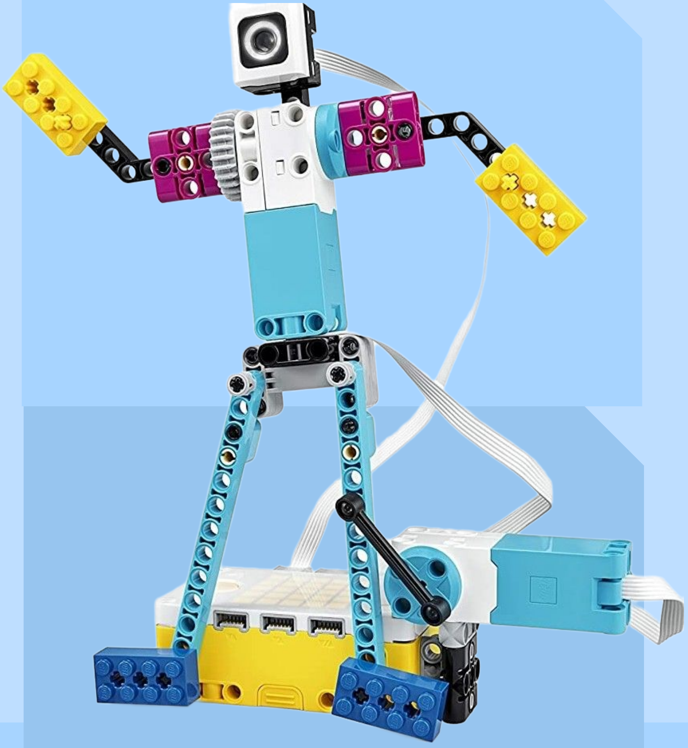
LESSON 2
Intro to Spike Prime

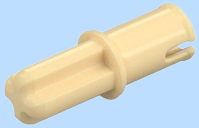


Spike Prime Materials

1 Pin
2 Types
Friction/Frictionless(Grey)

2 Pin
2 Types
Friction/Frictionless(Beige)

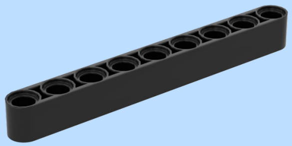
Pin/Axle Connector
2 Types
Friction/Frictionless(Beige)

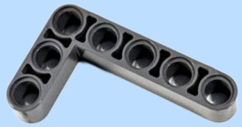


Spike Prime Materials

Beam
1 X (number of holes)
Only odd number of holes

Angled Beam
90,45,30



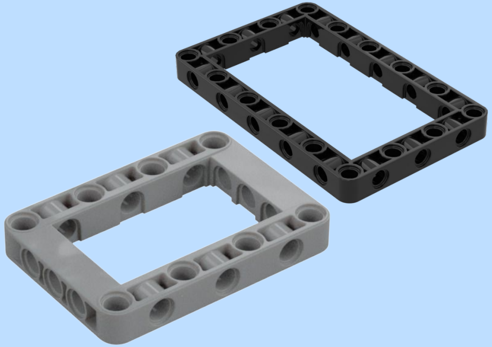


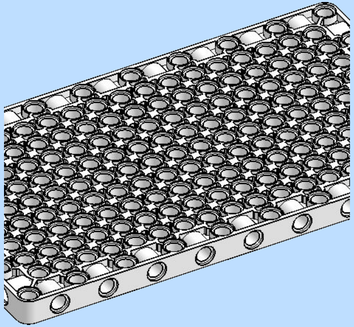
Spike Prime Materials

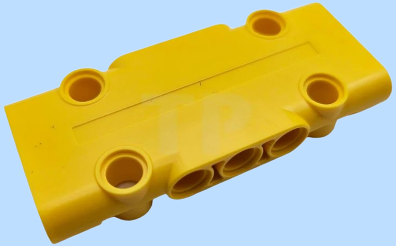
Picture Frame
Small, Med, Large

Panel
11x19

Plates
Small, Med, Large







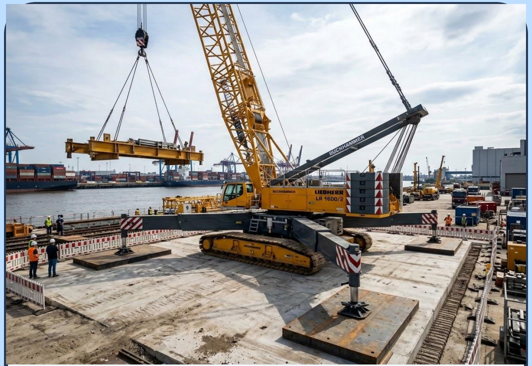
Engineering Words for Today

Strong

Can hold weight or force without breaking.

Rigid

Does not bend, wiggle, or twist easily.



Stable

Does not fall over easily.

Structural Forces: Tension & Compression

The forces that act on members are called tensile forces or compression forces.

Tensile Forces

These forces will **stretch** the structure.

Compression Forces

These forces will **squeeze** the structure.

Defining Structural Members

Ties

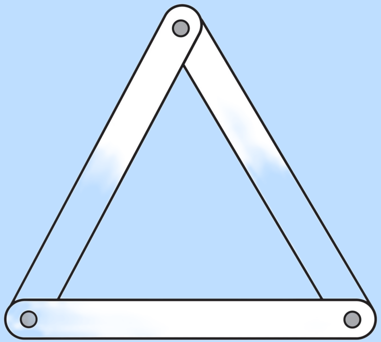
Members that are in **tension**.

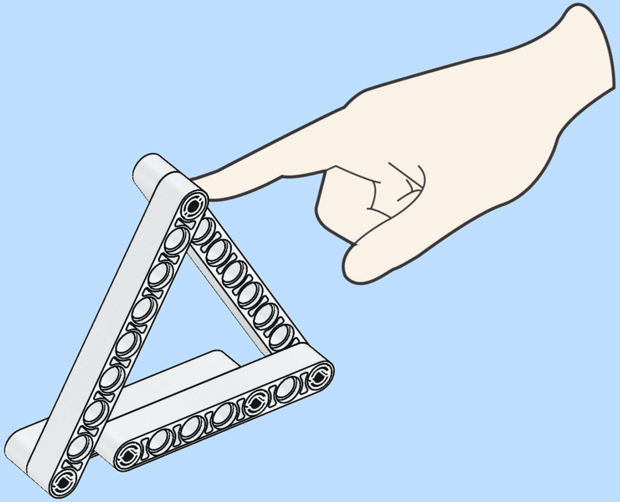
Struts

Members that are under **compression**.

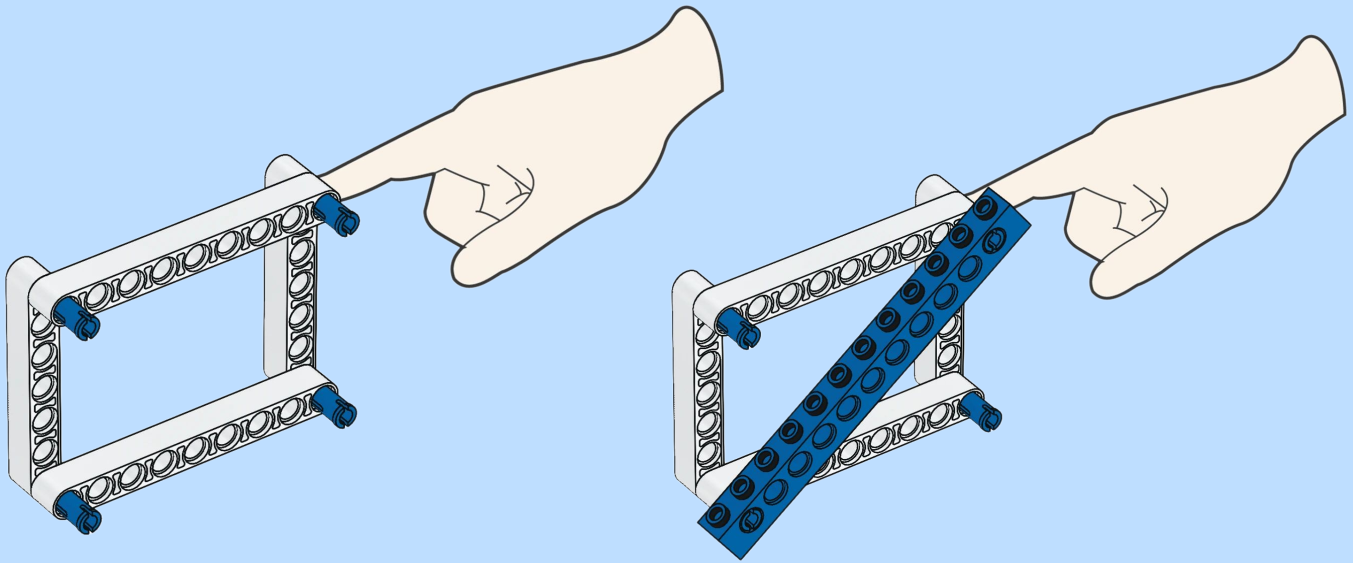
Building Strong Structures Tip 2

In bridges, cranes, towers and even space stations, triangulation is often used to make structures rigid.





A frame structure is made from pieces called members. This frame is rigid because it is triangulated.



Today's Challenge: Building a Bridge

Requirement: Create a bridge that is 14 inches apart

Testing Protocol

- Cup will be placed
- Marbles will be placed inside the cups

Completion

When the timer goes off:

Raise Your Hand